

Black-box language model explanation by context length probing

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Abstract

The increasingly widespread adoption of large language models has highlighted the need for improving their explainability. We present *context length probing*, a novel explanation technique for causal language models, based on tracking the predictions of a model as a function of the length of available context, and allowing to assign *differential importance scores* to different contexts. The technique is model-agnostic and does not rely on access to model internals beyond computing token-level probabilities. We apply context length probing to large pre-trained language models and offer some initial analyses and insights, including the potential for studying long-range dependencies. The source code¹ and an interactive demo² of the method are available.

1 Introduction

Large language models (LMs), typically based on the Transformer architecture (Vaswani et al., 2017), have recently seen increasingly widespread adoption, yet understanding their behaviour remains a difficult challenge and an active research topic.

Notably, as the length of the context that can be accessed by LMs has grown, a question that has attracted some attention is how this influences their predictions. Some recent studies in this line of research suggest that even “long-range” LMs focus heavily on local context and largely fail to exploit distant ones (O’Connor and Andreas, 2021; Sun et al., 2021; Press et al., 2021; Sun et al., 2022). A more nuanced understanding of how contexts of different lengths influence LMs’ predictions may hence be valuable for further improving their performance, especially on tasks like long-form text generation where long-range dependencies are of critical importance.

¹<https://github.com/cifkao/context-probing/>

²<https://cifkao.github.io/context-probing/>

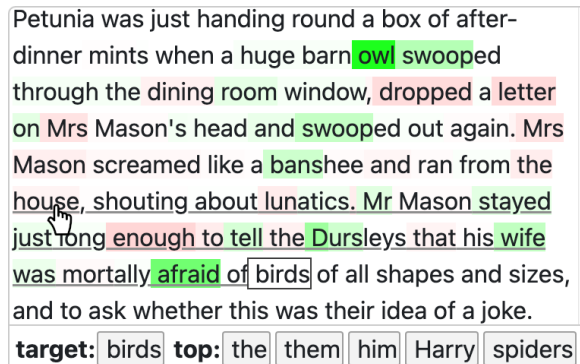


Figure 1: A screenshot of a demo² of the proposed method. After selecting a target token (here “**birds**”), the preceding tokens are highlighted according to their (normalized) *differential importance scores* (green = positive, red = negative), obtained using our method. The user can also explore the top predictions for contexts of different lengths (here the context “house, shouting about lunatics. [...] mortally afraid of”).

In this work, we propose *context length probing*, a simple explanation technique for *causal* (autoregressive) language models, based on tracking the predictions of the model as a function of the number of tokens available as context. Our proposal has the following advantages:

- It is conceptually simple, providing a straightforward answer to a natural question: *How does the length of available context impact the prediction?*
- It can be applied to a pre-trained model without retraining or fine-tuning and without training any auxiliary models.
- It does not require access to model weights, internal representations or gradients.
- It is model-agnostic, as it can be applied to any causal LM, including attentionless architectures like RNN (Mikolov et al., 2010) and CNN (Dauphin et al., 2017). The only requirement for the model is to accept arbitrary input

segments (i.e. not be limited to document prefixes).

Furthermore, we propose a way to use this technique to assign what we call *differential importance scores* to contexts of different lengths. This can be seen as complementary to other techniques like attention or saliency map visualization. Interestingly, contrary to those techniques, ours appears promising as a tool for studying long-range dependencies, since it can be expected to highlight important information not already covered by shorter contexts.

2 Related work

A popular way to dissect Transformers is by visualizing their attention weights (e.g. Vig, 2019; Hoover et al., 2020). However, it has been argued that this does not provide reliable explanations and can be misleading (Jain and Wallace, 2019; Serrano and Smith, 2019). A more recent line of work (Elhage et al., 2021; Olsson et al., 2022) explores “mechanistic explanations”, based on reverse-engineering the computations performed by Transformers. These techniques are tied to concrete architectures, which are often “toy” versions of those used in real-world applications, e.g. attention-only Transformers in Elhage et al.

Other options include general-purpose methods like neuron/activation interpretation (e.g. Geva et al., 2021; Goh et al., 2021; Dai et al., 2022), saliency maps (e.g. Fong and Vedaldi, 2017; Ancona et al., 2019) and influence functions (Koh and Liang, 2017). These require access to internal representations and/or the ability to backpropagate gradients, and have some caveats of their own (Kindermans et al., 2019; Kokhlikyan et al., 2021).

More closely related to our work are studies that perform *ablation* (e.g. by shuffling, truncation or masking) on different contexts to understand their influence on predictions (O’Connor and Andreas, 2021; Sun et al., 2021; Press et al., 2021; Vafa et al., 2021). To our knowledge, all such existing works only test a few select contexts or greedily search for the most informative one; in contrast, we show that it is feasible to consider *all* context lengths in the range from 1 to a maximum $c_{\max}$, which permits us to obtain fine-grained insights on the example level, e.g. in the form of the proposed differential importance scores. Moreover, many existing analyses (e.g. Vafa et al., 2021; O’Connor and Andreas, 2021) rely on specific training or fine-tuning, which is not the case with our proposal.

3 Method

3.1 Context length probing

A causal LM estimates the conditional probability distribution of a token given its left-hand context in a document:

$$p(x_{n+1} \mid x_1, \dots, x_n). \quad (1)$$

We are interested here in computing the probabilities conditioned on a *reduced* context of length $c \in \{1, \dots, n\}$:

$$p(x_{n+1} \mid x_{n-c+1}, \dots, x_n), \quad (2)$$

so that we may then study the behavior of this distribution as a function of c .

An apparent obstacle in doing so is that applying the model to an arbitrary subsequence $x_{n-c+1}, \dots, x_n$, instead of the full document $x_1, \dots, x_N$, may lead to inaccurate estimates of the probabilities in Eq. (2). However, we note that large LMs are not usually trained on entire documents. Instead, the training data is pre-processed by shuffling all the documents, concatenating them (with a special token as a separator), and splitting the resulting sequence into *chunks* of a fixed length (usually 1024 or 2048 tokens) with no particular relation to the document length. Thus, the models are effectively trained to accept sequences of tokens starting at arbitrary positions in a document and it is therefore correct to employ them as such to compute estimates of Eq. (2).³

It now remains to be detailed how to efficiently evaluate the above probabilities for all positions n and context lengths c . Specifically, for a given document $x_1, \dots, x_N$ and some maximum context length $c_{\max}$, we are interested in an $(N - 1) \times c_{\max} \times |\mathcal{V}|$ tensor $\mathbf{P}$, where $\mathcal{V} = \{w_1, \dots, w_{|\mathcal{V}|}\}$ is the vocabulary, such that:

$$\mathbf{P}_{n,c,i} = p(x_{n+1} = w_i \mid x_{n-c+1}, \dots, x_n), \quad (3)$$

with $\mathbf{P}_{n,c,*} = \mathbf{P}_{n,n-1,*}$ for $n \leq c$.⁴ Observe that by running the model on any segment $x_m, \dots, x_n$, we obtain all the values $\mathbf{P}_{m+c-1,c,*}$ for $c \in \{1, \dots, n - m + 1\}$. Therefore, we can fill in the tensor $\mathbf{P}$ by applying the model along a sliding window of size $c_{\max}$, i.e. running it on N (overlapping)

³For models trained on data that is pre-processed differently, (re)training or fine-tuning with data augmentation such as random shifts may be needed in order to apply our method, analogously to Vafa et al. (2021), who use word dropout to ensure compatibility with their method.

⁴ $\mathbf{P}_{n,c,*}$ is a $|\mathcal{V}|$ -dimensional slice of $\mathbf{P}$ along the last axis.

segments of length at most $c_{\max}$. See [Appendix A](#) for an illustration and additional remarks.

3.2 Metrics

Having obtained the tensor $\mathbf{P}$ as we have just described, we use it to study how the predictions evolve as the context length is increased from 1 to $c_{\max}$. Specifically, our goal is to define a suitable metric that we can compute from $\mathbf{P}_{n,c,*}$ and follow it as a function of c (for a specific n or on average).

One possibility would be to use the negative log-likelihood (NLL) loss values:

$$-\log p(x_{n+1} \mid x_{n-c+1}, \dots, x_n). \quad (4)$$

However, this may not be a particularly suitable metric for explainability purposes, as it depends (only) on the probability assigned to the *ground truth* x_{n+1} , while the LM outputs a probability distribution $\mathbf{P}_{n,c,*}$ over the entire vocabulary, which may in fact contain many other plausible continuations. For this reason, we propose to exploit a metric defined on whole *distributions*, e.g. the Kullback-Leibler (KL) divergence. To achieve this, we choose the maximum-context predictions $\mathbf{P}_{n,c_{\max},*}$ as a reference and get:

$$\begin{aligned} \mathcal{D}_{n,c} &= D_{\text{KL}}[\mathbf{P}_{n,c_{\max},*} \parallel \mathbf{P}_{n,c,*}] \\ &= \sum_{i=1}^{|\mathcal{V}|} \mathbf{P}_{n,c_{\max},i} \log \frac{\mathbf{P}_{n,c_{\max},i}}{\mathbf{P}_{n,c,i}}. \end{aligned} \quad (5)$$

The rationale for (5) is to quantify the amount of information that is lost by using a shorter context $c \leq c_{\max}$. Interestingly, this metric is *not* related to the absolute performance of the model with maximal context, but rather to how the output *changes* if a shorter context is used.

3.3 Differential importance scores

We are also interested in studying how individual *increments* in context length affect the predictions. We propose to quantify this as the change in the KL divergence metric (5) when a new token is introduced into the context. Specifically, for a pair of tokens x_{n+1} (the *target token*) and x_m (the *context token*), we define a *differential importance score* (Δ -score for short)

$$\Delta \mathcal{D}_{n,m} = \mathcal{D}_{n,n-m-1} - \mathcal{D}_{n,n-m}. \quad (6)$$

We may visualize these scores as a way to explain the LM predictions, much like is often done with

name	#param	#layer	#head	d_{model}	max len
gpt2	117 M	12	12	768	1024
gpt2-xl	1.5 B	48	25	1600	1024
gpt-j-6B	6.1 B	28	16	4096	2048

Table 1: Hyperparameters of the 3 models used.

attention weights, with two important differences. First, a high $\Delta \mathcal{D}_{n,m}$ should not be interpreted as meaning that x_m in isolation is important for predicting x_{n+1} , but rather that it is salient given the context that follows it (which might mean that it brings information not contained in the following context). Second, unlike attention weights, our scores need not sum up to one, and can be negative; in this regard, the proposed representation is more conceptually similar to a saliency map than to an attention map.

4 Results

We apply the proposed technique to publicly available pre-trained large Transformer language models, namely GPT-J ([Wang and Komatsuzaki, 2021](#)) and two GPT-2 ([Radford et al., 2019](#)) variants – see [Table 1](#) for an overview. We use the validation set of the English LinES treebank⁵ from Universal Dependencies (UD; [Nivre et al., 2020](#)), containing 8 documents with a total length of 20 672 tokens⁶ and covering fiction, an online manual, and Europarl data. We set $c_{\max} = 1023$. We use the 🤗 Transformers library⁷ ([Wolf et al., 2020](#)) to load the pre-trained models and run inference. Further technical details are included in [Appendix B](#).

4.1 LM loss by context length

[Fig. 2](#) shows the cross entropy losses (NLL means) across the whole validation dataset as a function of context length c . As expected, larger models perform better than smaller ones, which is traditionally explained by their larger capacity. A less common observation we can make thanks to this detailed representation is that the gains in performance come mostly from relatively short contexts (8–256 tokens); this is consistent with prior works ([Sun et al., 2021](#); [Press et al., 2021](#)) which found

⁵https://universaldependencies.org/treebanks/en_lines/index.html

⁶After concatenating all sentences and applying the GPT-2 tokenizer, which is used by both GPT-2 and GPT-J.

⁷<https://github.com/huggingface/transformers>